Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think> To find the de Broglie wavelength of an electron with a kinetic energy of 1 eV , we need to follow these steps :

1 . Convert the kinetic energy from eV to joules .

2 . Use the equation for kinetic energy to find the velocity .

3 . Use the velocity and mass of the electron to find the de Broglie wavelength .

1 eV = 1 . 602 ÃĹ 10 ^-19 J

Kinetic Energy (KE) = (1 / 2) * m * v ^ 2

where m is the mass of the electron (9 . 109 ÃĹ 10 ^-31 kg)

KE = 1 . 602 ÃĹ 10 ^-19 J

So , v = sqrt ((2 * KE) / m)

= sqrt ((2 * 1 . 602 ÃĹ 10 ^-19) / (9 . 109 ÃĹ 10 ^-31))

âīĪ 5 . 93 ÃĹ 10 ^ 6 m/s

de Broglie wavelength (Î») = h / (m * v)

where h is Planck's constant (6 . 626 ÃĹ 10 ^-34 Js)

Î» = (6 . 626 ÃĹ 10 ^-34) / (9 . 109 ÃĹ 10 ^-31 * 5 . 93 ÃĹ 10 ^ 6)

âīĪ 1 . 23 ÃĹ 10 ^-10 m or 1 . 23 ÃĹ 10 ^-7 cm (because 1 m = 100 cm)

Therefore , the de Broglie wavelength is 1 . 23 ÃĹ 10 ^-7 cm .

</think>

<answer> 1 . 23 ÃĹ 10 ^-7 cm </answer>